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SUPERSEDING
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1 August 1986

DETAIL SPECIFICATION

ROCKET MOTOR, MK 117 MOD 0

This specification is approved for use by the
Naval Air Systems Command, Department of the Navy, and is available
for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes requirements for the Mk 117 Mod 0 Rocket Motor (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Code 414100B120-3, Highway 547, Lakehurst, NJ 08733-5100, by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-453	-	Inspection, Radiographic
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines
MIL-STD-1167	-	Ammunition Data Card
MIL-STD-1168	-	Lot Numbering of Ammunition
DOD-STD-2101	-	Classification of Characteristics

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Bldg 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE CODE 30003)

DL1638AS120	-	Rocket Motor, Mk 117 Mod 0

(Copies of this drawing are available from the Indian Head Division, Naval Surface Warfare Center, Data Review and Control Branch, Code 8410P, 101 Strauss Ave., Indian Head, MD 20640-5035.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in the document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

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3.2 Conformance to documents. The rocket motor shall conform to this specification and the documents listed on DL1638AS120.

3.3 Characteristics.

3.3.1 Radiographic innspection (M101). All rocket motor assemblies shall be radiographically inspected (see 4.5.2).

3.3.2 Circuit resistance (M102). The igniter circuit resistance on all rocket motor assemblies, when measured from the connector at 70° ± 10°F, shall be between 0.45 to 0.75 ohms (see 4.5.3).

3.4 Ballistic requirements (M103). Rocket motors shall meet the ballistic requirements in table I over the temperature range of -40°F to 140°F (see 4.5.5).

Table I. Ballistics requirements.

Parameter	Requirement
Total impulse (lb f-s)	3400 min
Action time (s)	2.5 max
Thrust (lb f)	3750 max
Ignition delay (s)	0.085 max

3.5 Workmanship. Motors and components, when examined visually in 4.5.1, shall be free from dents, scratches, split or cracked components, poor quality welds, chippping or flaking of plated or painted surfaces, sharp edges or other defects or discontinuities as specified on the applicable drawings.

3.6 Data cards. Data cards, when completed in accordance with MIL-STD-1167, shall be included for each production lot of rocket motors.

3.7 Lot numbering. Lot numbering shall be in accordance with MIL-STD-1168.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Conformance inspection (see 4.4)

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the applicable test method of 4.5.

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4.3 First article inspection. The first article sample shall consist of 18 rocket motors. The sample shall undergo the test and inspections specified in table II. Each rocket motor shall be visually inspected and undergo circuit resistance testing after each test specified in table II. Failure of any rocket motor shall be cause for first article rejection.

4.4 Conformance inspection. Conformance inspection shall consist of the test and inspections in table III.

4.4.1 Lot formation. A lot shall consist of all the rocket motor units loaded from the same propellant grain lot and the same igniter lot and submitted for inspection and test at the same time.

4.4.2 Sampling.

4.4.2.1 Nondestructive tests. Unless otherewise specified (see 6.2), all motors shall be nondestructively inspected in accordance with table III. There shall be no defects.

4.4.2.2 Destructive test. Destructive tests shall be the ballistic tests in accordance with table III. The sample for destructive tests shall be selected at random from each lot as determined by the acquiring activity for each temperature tested.

4.4.3 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as critical, major, or minor in accordance with DOD-STD-2101. Requirements that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the requirement. Requirements not annotated with a classification code are classified minor.

4.5 Test methods

4.5.1 Visual. Motors and components shall be visually inspected to ensure the requirements of 3.5 are met.

4.5.2 Radiographic inspection (M101). Each rocket motor assembly shall be subjected to radiographic inspection in accordance with MIL-STD-453, quality level I (see 3.3.1).

4.5.3 Circuit resistance (M102). The rocket motor igniters shall be tested for circuit resistance by applying 0.010 amperes (maximum) to the initiator assembly circuit at 70° ± 10°F. Compliance to 3.3.2 shall be met.

4.5.4 Environmental and operational tests.

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Table II. First article inspection.

Treatment or test	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	Test paragraph
Visual	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	4.5.1
X-ray and circuit resistance	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	4.5.2, 4.5.3
High temperature storage									x	x	x	x	x	x	x	x	x	x	4.5.4.1
X-ray and circuit resistance									x	x	x	x	x	x	x	x	x	x	4.5.2, 4.5.3
Temperature shock											x	x	x	x	x	x	x	x	4.5.4.2
X-ray and circuit resistance											x	x	x	x	x	x	x	x	4.5.2, 4.5.3
Salt and fog											x	x							4.5.4.3
Vibration																			4.5.4.4
@ -40°F											x				x		x		
@ 140°F												x				x		x	
X-ray and ciruit resistance											x	x			x	x	x	x	4.5.2, 4.5.3
Two-foot drop											x	x							4.5.4..5
X-ray and circuit resistance											x	x							4.5.2, 4.5.3
Acceleration firings							x	x											4.5.4.6
Ballistic test																			4.5.5
@ -40°F	x	x	x						x		x		x		x	x			
@140°F				x	x	x				x		x		x			x	x	

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4.5.4.1 High temperature storage. Rocket motors shall be placed in a conditioning chamber at 140° F for 45 days in accordance with MIL-STD-810, Method 501.3, Procedure I

Table III. Conformance inspection.

Inspection or test	Requirement paragraph	Test paragraph
Nondestructive inspection		
Visual	3.5	4.5.1
Radiographic	3.3.1	4.5.2
Circuit resistance	3.3.2	4.5.3
Destructive inspection		
Ballistic test	3.4	4.5.5
@ -40°F		
@ 140°F		

4.5.4.2 Temperature shock The rocket motors shall withstand temperature shock in accordance with MIL-STD-810, Method 503.3. The motors shall be equally divided into two groups with each group subjected to alternate conditioning chambers at -40°F and 140°F. One group shall begin and end at 40°F, the other at 140°F. The motors shall remain in each chamber for not less than 8 hours, then transferred within 5 minutes to the other chamber. Each motor shall undergo a total of 3 cycles (see 6.5.6). After temperature cycling, the grain shall be returned to ambient temperature and radiographically inspected as specified in 4.5.2.

4.5.4.3 Salt and fog. The rocket motor sahll be subjected to an aqueous salt atmosphere in accordance with Procedure I, Aggravated Screening, Method 509.3 of MIL-STD-810. Test duration shall be 48 hours. After exposure, the motor shall be allowed to dry for 48 hours under ambient conditions.

4.5.4.4 Vibration. The rocket motor shall be subjected to Category I, Basic Transportation Vibration, Method 514.4 schedules defined by Figures 514.4-1 through 514.4-3 of MIL-STD-810. Expected transportation distance shall be 3000 miles and the rocket motor shall be treated as secured cargo. The rocket motor shall be maintained at the temperature specified in table II.

4.5.4.5 Two-foot drop. The uncrated rocket motors shall withstand a drop from 2 feet, measured from the lowest point on the motor to the impact surface. The impact surface shall be a 2-inch thick steel plate (Brinell hardness of not less than 207) embedded in not less than 2 feet of reinforced concrete. Motors shall be dropped horizontally. Motors shall be temperature conditioned for not less than 8 hours at -40°F before testing. Motors shall be drop tested within 15 minutes after removal from conditioning. If out of conditioning for more than 15 minutes, the motors shall be reconditioned at -40°F. Reconditioning time shall be 2 minutes for each minute

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(over 15) the motor was out of the conditioning chamber. Reconditioning time shall be not more than 8 hours.

4.5.4.6 Acceleration. During firings, the rocket motor shall withstand a simulated acceleration of 15 g's through the longitudinal axis in accordance with MIL-STD-810, Method 513.4.

4.5.5 Ballistic test (M103). Rocket motors shall be ballistically tested in accordance with 4.5.5.1 and 4.5.5.2 (see 3.4).

4.5.5.1 Test setup. The rocket motor shall be mounted on a suitable test stand, equipped with a recording oscillograph and instrumentation for recording thrust and time. The motor shall be fired not later than 15 minutes after removal from the conditioning chamber. If this time limit is exceeded, the motors shall be reconditioned 2 minutes for every minute over 15 that the out of specification condition exists, or not more than 8 hours. Motors shall be fired by energizing the igniter circuit with 28 volts dc at 7 +0.5, -0.0 amperes.

4.5.5.2 Instrumentation. All measurements shall be made with instruments and systems with a total system accuracy, reproducibility, and frequency response that has been approved by the acquiring activity.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The rocket motors are intended to provide auxiliary thrust to launch, from either ground or ship, the BQM-74C target drone.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DoDISS to be cited in the solicitation and, if required, the specific issue of

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individual documents referenced (see 2.2.1).

c. Packaging requirements (see 5.1 and 6.4).

d. Mark and Mod of the rocket motor.

e. Whether first article is required (see 4.3 and 6.3).

f. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M are applicable.

g. Inspection conditions, if other than as specified (see 4.2).

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of 18 rocket motors. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and dispostion of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first aricle inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Packaging. Contracting officers are advised that the preferred packaging instructions for the Mk 117 Mod 0, rocket motor are contained in Drawing 1638AS123, "Packing and Marking, Rocket Motor, Mk 117 Mod 0."

6.5 Definitions.

6.5.1 Thrust-time curve for rocket motor, Mk 117 Mod 0 (see figure 1).

6.5.2 Action time. Action time is the time between the initial 200-pound thrust level and the final 200-pound thrust level.

6.5.3 Total impulse. Total impulse is the integral of thrust over the action time.

6.5.4 Ignition delay. Ignition delay is the time from zero time (moment current is applied to igniter) to the initial 200-pound thrust.

6.5.5 Maximum thrust. The maximum thrust is the highest thrust attained by the motor, after 100ms, during a static-firing test, recorded to the nearest lbf.

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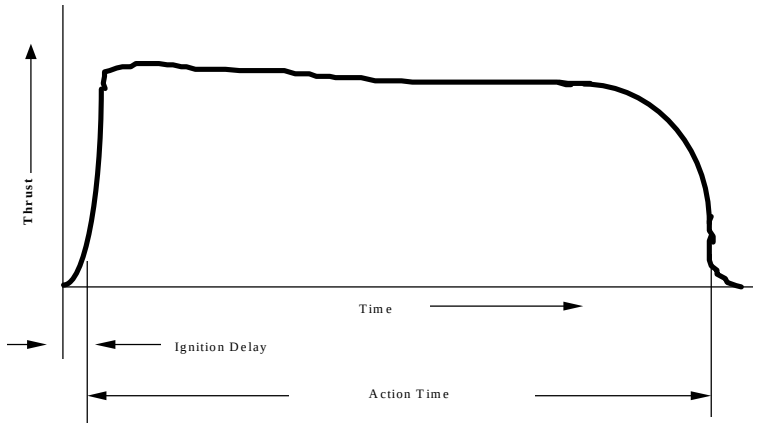


Figure 1. Sample thrust-time curve.

6.5.6 Cycle. A cycle is defined as -40°F to 140°F and back to -40°F, or alternatively from 140°F to -40°F and back to 140°F.

6.6 Subject term (key word) listing.

Ignition delay
Thrust
Total impulse

6.7 Changes from the previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing activity
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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-DTL-85769A(AS)	2. DOCUMENT DATE (*YYMMDD*)96/6/21

3. DOCUMENT TITLE ROCKET MOTOR, MK117 MOD 0

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial: (2) DSN: (*If Applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a. NAME COMMANDER NAVAL AIR WARFARE CENTER AIRCRAFT DIVISION	b. TELEPHONE NUMBER (*Include Area Code*) (1) Commercial (908) 323-7488 (2) DSN 624-7488
c. ADDRESS (*Include Zip Code*) CODE 4.1.11.2B120-3 HIGHWAY 547 LAKEHURST, NJ 08733-5100	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office, 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

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